OneMath

Created by: Hector Lee

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Background

Currently, as a second-year volunteer, I believe that there is an objective-mismatch between parents/caregivers, and SHG/CDAC. Although it has been emphasised to the caregivers that SHG is not a tuition program, I believe that they still have the primary aim that their children's academic competency will improve under this program. I base this belief on two factors:

1. Communication during the introductory session in week 1. The parents/caregivers had a focus on the academic aspects of the program, ranging from questions regarding the duration of breaks taken, to the competency level of the volunteers with regards to the academic coaching. Conversely, there was a lack of queries about the activities from them.
2. The fact that the students are upper primary, specifically, P5 and/or P6. As of right now, the most important event in their academic journey currently would be their PSLE. When viewing the opportunity cost incurred by the 3-hour session, I believe that concluding that the parents/caregivers would want their children to spend time fruitfully towards academic activities that would contribute to their PSLE grades is natural.

Therefore, the parents/caregivers who send their children/wards to SHG believe that the program will boost their academic results (though the exact extent is unknown and varies from individual to individual).

However, SHG/CDAC still has the aim that SHG should not be a "tuition-program", which is contradictory to what the parents want.

In lieu of this objective-mismatch, I propose a technological solution to improve the grades of the students without compromising SHG's aim.

Introduction

Speaking as a volunteer, I believe that the main reason why the students have poorer academic capabilities than their abilities suggest is due to <u>a lack of practice</u>. <u>Despite the upcoming year-end PSLE, most of their practice stems from school, SHG sessions, and tuition lessons (if applicable); they lack consistent, daily self-practice at home.</u>

I propose a technological solution to solve the lack of practice. Specifically, focusing on mathematics, a simple web application that gives the student a topic-based math question to do daily. These math questions will be sourced and adapted from past PSLE questions and other primary school exams. The aim of the web application will be to get the student to answer a question correctly. Should the student get the question wrong, another question will be given until he answers it correctly. In the meantime, the topic he made a mistake in will be recorded, so that the next time the student asks for a question, the higher the chance that a question of that topic will be given.

Details

All images are obtained from a currently working prototype

- Each student will have a unique profile without any Personal Identifiable Information (PII); each student will have a unique username to identify and track their progress with. See Figure 1

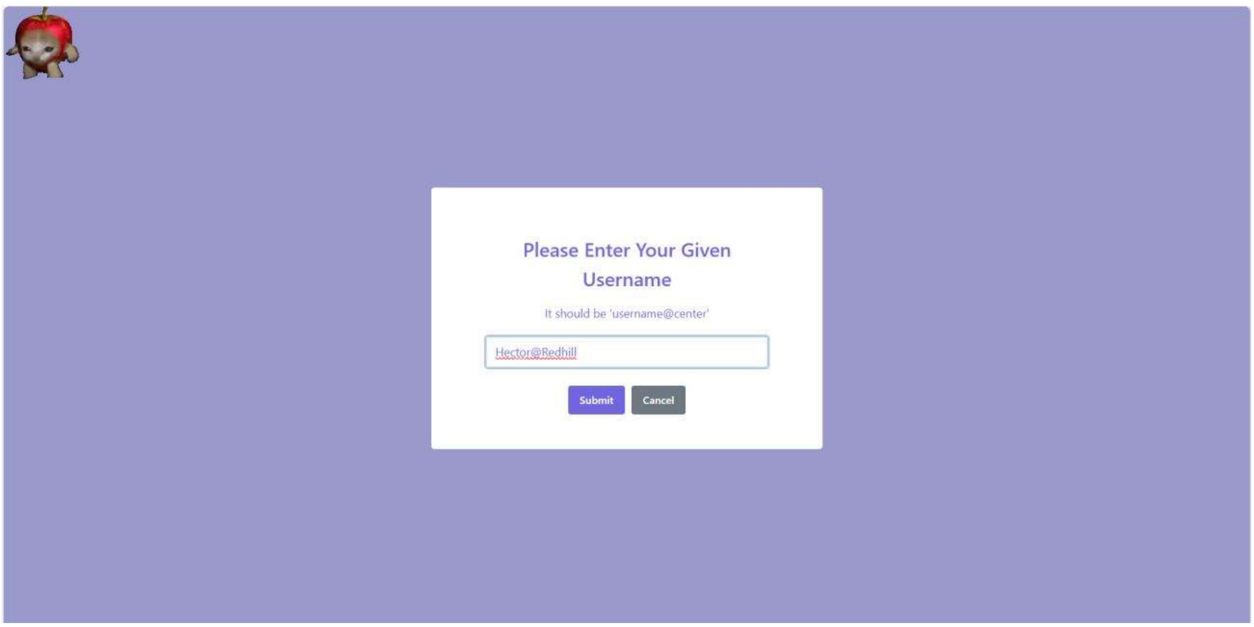

Figure 1: Unique Username

- Questions will be segregated by topics stipulated by MOE's PSLE syllabus document, with answers and workings being submitted by volunteers on a one-off basis to be input into the database. See Figure 2 for an example question

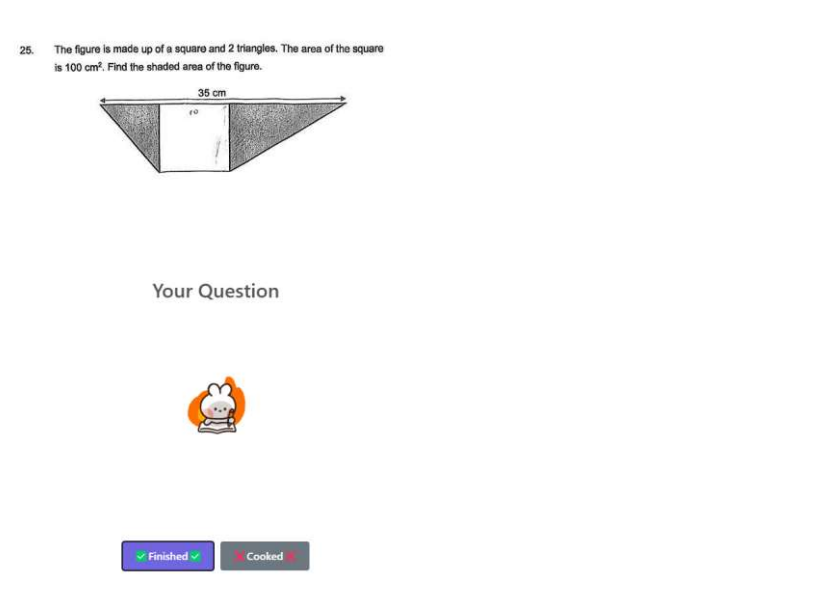

Figure 2: An example question

- When a student gets a question, should he get it correct, he has the option of continuing on. Should he get it wrong, the working and solution will be shown, and another question will be prompted until he answers one correctly.

- Each student will have a list of scores based on topic. Each score will start at 0. Should the student get the question wrong, 1 will be deducted from his score. Conversely if he gets it correct, 1 will be added to his score. Therefore the next day when the student gets a question, the system will give him a question based on his topics in which he has a negative score in, with the aim of making all his scores for each topic 0. Should he all have positive scores, a random question will be given instead.

Benefits

I believe that this solution will be helpful to students without compromising SHG's primary goal.

Students will be able to obtain personalised practice at home on a daily-basis, so that they can improve. In addition, should they have any issues with the questions, they can note it down and bring it to SHG sessions to discuss with their volunteers, providing a more guided approach towards academics in SHG sessions, instead of it turning into a "homework completion session" where students bring their schoolwork to get it done in SHG, only going through the motions.

Simultaneously, SHG's primary goal of not being a specialised tuition program is not compromised since the dual emphasis on academics and activities is still present. This web application merely serves as a supplement to students so that they can practice more frequently.

Areas For Improvement

1) Students might lack perseverance to persist in daily practice. I believe that given cooperation between centers, parents, and volunteers, the goal of only 1 correct question a day can be achieved.

2) Man-hours must be put in into ensuring that the question-and-answer base is robust. Although volunteers will have to physically do the questions and screenshot both questions and answers themselves to submit into the database, this is only a one-off occurrence because the data will be persisted into the database. Therefore, the same question-and-answer bank can be used for new batches.

3) Scoped project focus on mathematics. Unfortunately, having been through PSLE, this approach will only work on mathematics and possibly science. To do well in languages, frequent reading of books will be needed.

Costs Incurred

The only costs involved would be sourcing for question-material, and hosting of the database and website.

With the presence of online examination papers, question-material costs should be low at the start. Additionally, should there be a need, used copies can be obtained since only the question is needed; answers will be provided by volunteers, providing one way to reduce future costs.

The database and website will be hosted for free since I will be hosting it. I plan on volunteering consistently with CDAC/SHG for the foreseeable future. In the unlikely event that my plans change, the website cost will be low since only a domain name needs to be obtained (which goes from SGD15-20 a year). As the only transactions in the web application will be retrieval of question-and-answer images, and verification of username when logging in, a low-cost service and database are needed. Additionally, the current prototype has taken measures to keep images below 20kb, with a current average of 17.5 kb. Therefore, 2GB of storage can store between 104,000 to 119,000 images, far more than realistically obtainable. Cheap hosting platforms like Heroku have pricings lesser than 7SGD a month, which can be additionally subverted should there be separate owned machines to run the database on, which is dockerised for convenience.

Conclusion

I believe that this web application will be able to achieve the primary goal of the parents (which is to improve the grades of their wards) without compromising the aim of CDAC/SHG (becoming a specialised tuition program), all by providing more opportunities for practice. Although students who use this app might not be able to get AL1, however, they could improve to consistently get an AL3 or an AL4.